

- **SAMPLING DISTRIBUTION: THE DISTRIBUTION OF SOME SAMPLE STATISTIC**
- **STATISTIC: ANY FUNCTION OF MY DATASET**

Lecture 3

MAIN RESULTS FROM LAST TIME

- $E(\bar{X}) = \mu$ ← THE EXPECTED VALUE OF THE ARITHMETIC MEAN IS EQUAL TO THE POP. MEAN
- $V(\bar{X}) = \frac{\sigma^2}{m}$ → POP. VARIANCE / SAMPLE SIZE
- $E(S^2) = \sigma^2$ → POP. VARIANCE

FROM $E(S^2) = \sigma^2$, WE CAN SAY $E(S) \leq \sigma$ → NOT EXACTLY EQUAL TO σ BUT IT IS LESS THAN σ
 ⇒ S AND S^2 ARE RELATED TO EACH OTHER WITH A NON-LINEAR RELATIONSHIP

A CONSEQUENCE OF THE JENSEN'S INEQUALITY $\varphi(E(X)) \leq E[\varphi(X)]$ WHEN φ CONVEX

WITH $\varphi(S) = S^2$, $E(S)^2 \leq E(S^2) = \sigma^2$

(SQUARE ROOT ON BOTH SIDES)
 $\Rightarrow E(S) \leq \sigma$

ALSO: $V(S^2) = \frac{\sigma^4}{m} \left(\beta_2 + 2 \frac{m}{m-1} \right)$, WHERE

VARIANCE OF SAMPLE VARIANCE
 SAMPLE VARIANCE IS A RANDOM VARIABLE

$\beta_2 = \frac{\mu_4}{\sigma^4} - 3$ EXCESS KURTOSIS

CURTOSIS OF THE GAUSSIAN DISTRIBUTION

"BETA 2"

FOURTH-CENTRAL MOMENT
 $\mu_4 = E((X-\mu)^4)$

SPECIAL CASE: $X \sim N(\mu, \sigma^2)$

$\Rightarrow \beta_2 = 0$ ← BECAUSE NORMAL DISTRIBUTION

$\Rightarrow V(S^2) = \frac{2\sigma^4}{m-1}$ (m CANCEL OUT)

EXAMPLE: A POP OF $N=8$ STATISTICAL UNIT: FAMILY

X : NUM OF COMPONENTS. IN THE POP: $FAM_1, FAM_2, \dots, FAM_8$

1	2	3	4	5	6	7	8
↓	↓	↓	↓	↓	↓	↓	↓
2	5	4	3	3	4	4	4

EXPRESS THE WHOLE SUPPORT OF RANDOM VARIABLE X

POSSIBLE VALUES OF X

$X = \begin{cases} 2 \\ 3 \\ 4 \\ 5 \end{cases}$ u.p. u.p. u.p. u.p.

ONE ELEMENT

TWO ELEMENTS

FOUR ELEMENTS

ONE ELEMENT

1

FAMILY LABEL

NUM. MEMBERS

WE KNOW THE ENTIRE POP. USUALLY THIS IS NOT THE CASE

POP. MEAN = SUM OF $X \cdot \text{PROB OF } X$

$\mu = \sum_{x \in \text{supp}(X)} x \cdot P(X=x)$

$= 2P(X=2) + \dots + 5P(X=5)$

$= 2 \cdot \frac{1}{8} + \dots + 5 \cdot \frac{1}{8} = 3.625$

USUALLY WE DO NOT OBSERVE BECAUSE WE DO NOT HAVE THE WHOLE POPULATION

POP. VARIANCE

$$\begin{aligned}\sigma^2 &= E(X^2) - E(X)^2 \\ &= (2^2 \cdot P(X=2) + \dots + 5^2 \cdot P(X=5)) - (3.625)^2 \\ &= \left(4 \cdot \frac{1}{8} + \dots + 25 \cdot \frac{1}{8}\right) - (3.625)^2 = 0.734\end{aligned}$$

INDEPENDENT
IDENTICALLY
DISTRIBUTED
(IID SAMPLES)

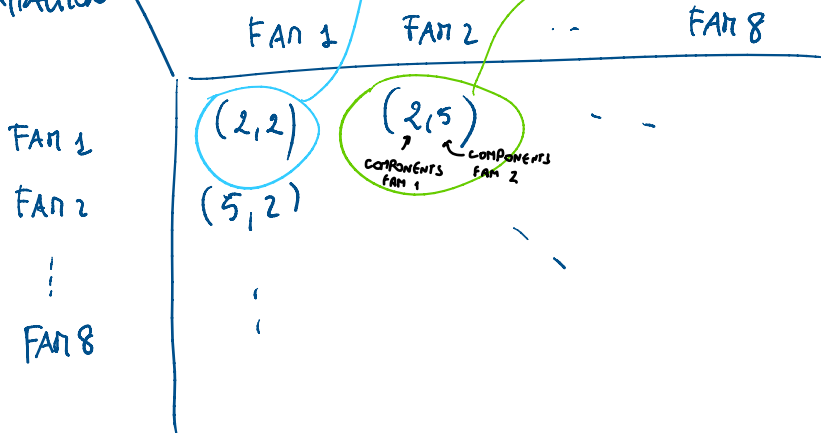
NOW: WE EXTRACT A RANDOM SAMPLE OF SIZE $M=2$

WHAT IS Ω (SAMPLE SPACE)?

2 EXTRACTION ($N=2$)

1ST EXTRACTION

2ND EXTRACTION



WITH REPLACEMENT

ONE POSSIBLE SAMPLE
I CAN OBSERVE

IN REALITY I
HAVE 1 REALIZED
 $\underline{X} = (X_1, X_2)$

WE CAN DERIVE THE DISTRIB. OF $\underline{X} = (X_1, X_2)$:

$X_1 \backslash X_2$	FAN 1, FAN 1				ROW SUMS
	2	3	4	5	
2	$\frac{1}{64}$ $\bar{X}=2$	$\frac{2}{64}$ $\bar{X}=2.5$	$\frac{4}{64}$ $\bar{X}=3$	$\frac{1}{64}$ $\bar{X}=3.5$	$\frac{8}{64} = \frac{1}{8}$
3					$P(X_1=3) = \frac{2}{8}$
4					$P(X_1=4) = \frac{4}{8}$
5					$P(X_1=5) = \frac{1}{8}$
COL SUMS	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{4}{8}$	$\frac{1}{8}$	

FAVORABLE OUTCOMES POTENTIALLY WITH $(X_1, X_2) = (2, 3)$:

- FAN 1, FAN 3
- FAN 1, FAN 5

$P(X_1=2)$ X_2 DOES NOT ENTER IN THIS PROB, IT COULD BE ANYTHING

MARGINAL DISTRIBUTION OF X_1

MARG. DISTR. OF X_2

ARE THE SAME, AS EXPECTED FROM i.i.d.

FROM JOINT DISTRIB. OF $\underline{X}$, WE CAN DERIVE THE
SAMPLING DISTR. OF $\bar{X} = \frac{X_1 + X_2}{2}$

ALL I WANT ARE THE PROBABILITIES
ASSOCIATED TO $\bar{X}$ (WE KNOW
FROM ABOVE)

POSSIBLE
VALUES FOR
ANY SAMPLE SIZE
(NOTES) OF 2
...

$$\bar{X} = \begin{cases} 2, & \text{u.p. } 1/64 \quad (\text{when } (X_1, X_2) = (2, 2) \text{ and } P(X = (2, 2)) = 1/64) \\ 2.5, & \text{u.p. } 4/64 \quad (\text{when } \underline{X} = (2, 3) \text{ u.p. } 2/64 \\ & \quad \underline{X} = (3, 2) \text{ u.p. } 2/64) \\ 3, & \\ 3.5, & \\ 4, & \\ 4.5, & \\ 5, & \text{u.p. } 1/64 \quad (\underline{X} = (5, 5) \text{ u.p. } 1/64) \end{cases}$$

$$\Rightarrow E(\bar{X}) = 2 \cdot \frac{1}{64} + 2.5 \cdot \frac{4}{64} + \dots + 5 \cdot \frac{1}{64} = 3.625 = \mu$$

$$V(\bar{X}) = E(\bar{X}^2) - E(\bar{X})^2 = \left((2)^2 \frac{1}{64} + (2.5)^2 \frac{4}{64} + \dots \right) - (3.625)^2 = 0.3672$$

$$= \frac{0.734}{2} = \frac{\sigma^2}{2}$$

DISTRIB. OF $\bar{X}$ IN SPECIAL CASES

• BERNOULLI DISTRIB. $X \sim \text{Bernoulli}(p) \Rightarrow \sum_{i=1}^m X_i \sim \text{Bin}(m, p)$

$(X_1, \dots, X_m)$ IID SAMPLE

$$\Rightarrow \text{I can compute } P\left(\sum_{i=1}^m X_i = t\right) = \binom{m}{t} p^t (1-p)^{m-t}, \quad t=0, 1, \dots, m$$

SPECIFIED VALUE

$$\text{THEN } P(\bar{X} = \bar{x}) = P(m\bar{X} = m\bar{x}) = P\left(\sum_{i=1}^m X_i = m\bar{x}\right) = \binom{m}{m\bar{x}} p^{m\bar{x}} (1-p)^{m-m\bar{x}}$$

EX:

$$p=0.3 \Rightarrow X \sim \text{Bern}(0.3) \Rightarrow \sum_{i=1}^5 X_i \sim \text{Bin}(5, 0.3)$$

$5 \cdot \frac{1}{m} \sum_{i=1}^m X_i$

$$P(\bar{X}=0) = P(m\bar{X} = m \cdot 0) = P\left(\sum_{i=1}^5 X_i = 0\right) = \binom{5}{0} (0.3)^0 (0.7)^{5-0} \approx 0.168$$

$$P(\bar{X}=0.2) = P(5 \cdot \bar{X} = 5 \cdot 0.2) = P\left(\sum_{i=1}^5 X_i = 1\right) = \binom{5}{1} (0.3)^1 (0.7)^{5-1} \approx 0.36$$

... SEE NOTES FOR THE WHOLE DISTRIB. OF $\bar{X}$.

• NORMAL DISTRIBUTION $X \sim N(\mu, \sigma^2)$

WE USE THE RESULT FOR WHICH IF $X_i \sim N(\mu_i, \sigma_i^2)$ i.i.d.

THEN ANY LINEAR COMBINATION OF NORMALS IS NORMAL:

$$\sum_{i=1}^m \boxed{a_i} X_i \sim N \left(\overset{\substack{\text{MEAN: LINEAR} \\ \text{COMBINATION OF MEAN}}}{\sum_{i=1}^m a_i \mu_i}, \sum_{i=1}^m a_i^2 \sigma_i^2 \right)$$

↑
KNOWN
CONSTANTS

$\Rightarrow \bar{X}$ IS A SPECIAL LINEAR COMBINATION WHEN $\begin{cases} \mu_i = \mu \\ \sigma_i^2 = \sigma^2 \\ a_i = 1/m \end{cases}$

$$\sum_{i=1}^m a_i X_i = \sum_{i=1}^m \frac{1}{m} X_i = \bar{X}$$

$$\frac{1}{m} \mu \sum_{i=1}^m 1 = \frac{1}{m} \mu \cdot m = \mu$$

$$\text{AND } \bar{X} \sim N \left(\sum_{i=1}^m \frac{1}{m} \mu, \sum_{i=1}^m \left(\frac{1}{m} \right)^2 \sigma^2 \right)$$

$$\frac{1}{m^2} \sigma^2 \cdot \sum_{i=1}^m 1 = \sigma^2 / m$$

$$\boxed{\bar{X} \sim N(\mu, \sigma^2 / m)}$$

NOTE: WHEN $X \sim$ UNKNOWN BUT m "LARGE",

THIS RESULT IS STILL VALID ASYMPTOTICALLY ($m \rightarrow \infty$)

THANKS TO CLT ^{CENTRAL LIMIT THEOREM}

$$\boxed{\bar{X} \approx N(\mu, \sigma^2 / m), m \text{ LARGE}}$$

WHEN $X \sim N(\mu, \sigma^2)$, WE CAN ALSO RECOVER THE DISTRIB. OF

A FUNCTION OF S^2 :

$$\boxed{(m-1) \frac{S^2}{\sigma^2} \sim \chi_{m-1}^2} \quad (\text{PROOF OMITTED})$$

$$(m-1) \frac{S^2}{\sigma^2} = \cancel{(m-1)} \cdot \frac{1}{\cancel{m-1}} \sum_{i=1}^m (X_i - \bar{X})^2 / \sigma^2 = \sum_{i=1}^m \left(\frac{X_i - \bar{X}}{\sigma} \right)^2$$

SUM OF
SQUARED
NORMALS

AND YOU HAVE χ_n^2 WHENEVER $\sum_{i=1}^n Z_i^2$, $Z_i \sim N(0,1)$

YOU CAN FIND QUANTILES OF χ_n^2 WITH

THEY WILL BE USEFUL WHEN

WE TALK OF CI AND HT ON σ^2

$\left\{ \begin{array}{l} \text{qchisq (in R)} \\ \text{chi2inv (in MATLAB)} \\ \text{scipy.stats.chi2.ppf (in Python)} \\ \text{chiinv (Excel)} \end{array} \right.$

CONF.
INTERVAL

HYPOTHESIS
TEST

STILL WHEN $X \sim N(\mu, \sigma^2)$, WE HAVE

$$\boxed{\bar{X} \perp S^2}$$

↑ STOCHASTIC
INDEPENDENCE
(JOINT = $\prod_i$ MARGINALS)